

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/232713208>

Towards green data centers: A comparison of x86 and ARM architectures power efficiency

Article in Journal of Parallel and Distributed Computing · December 2012

DOI: 10.1016/j.jpdc.2012.08.005

CITATIONS

88

READS

5,985

2 authors:



[Rafael V Aroca](#)

Federal University of São Carlos

52 PUBLICATIONS 668 CITATIONS

[SEE PROFILE](#)



[Luiz Marcos G. Gonçalves](#)

Federal University of Rio Grande do Norte

287 PUBLICATIONS 2,652 CITATIONS

[SEE PROFILE](#)



Towards green data centers: A comparison of x86 and ARM architectures power efficiency



Rafael Vidal Aroca*, Luiz Marcos Garcia Gonçalves

Federal University of Rio Grande do Norte, Computing Engineering and Automation Department, Technology Center, NatalNet Laboratory, 59078-900 - Natal - RN, Brazil

ARTICLE INFO

Article history:

Received 22 October 2011

Received in revised form

22 August 2012

Accepted 24 August 2012

Available online 5 September 2012

Keywords:

Low power

Cluster

Power usage

Performance evaluation

Energy efficiency

ABSTRACT

Servers and clusters are fundamental building blocks of high performance computing systems and the IT infrastructure of many companies and institutions. This paper analyzes the feasibility of building servers based on low power computers through an experimental comparison of server applications running on x86 and ARM computer architectures. The comparison executed on web and database servers includes power usage, CPU load, temperature, request latencies and the number of requests handled by each tested system. Floating point performance and power usage are also evaluated. The use of ARM based systems has shown to be a good choice when power efficiency is needed without losing performance.

© 2012 Elsevier Inc. All rights reserved.

1. Introduction

Much has been said about green computing and energy efficient data centers recently, but most solutions are focused in enhancing current server architectures or using virtualization [20] and other techniques to consume less power in data centers. This article discusses if processors typically used in mobile devices are suitable for server and cluster applications. Considering such possibilities, we designed a series of experiments in order to compare x86 and ARM systems in the context of two classical network services applications. Several hardware platforms are evaluated, measuring their power usage, temperature and application performance. We also analyze floating point performance and its power usage using Linpack [6].

Low power consumption is historically related to embedded systems running with batteries, but over time more data centers and high performance systems are concerned with power efficiency [19,2]. One characteristic of data centers is their huge electric power consumption [21], which is a serious problem [16]. According to Mahadevan et al., the biggest consumers of energy are servers and cooling systems [21]. Reducing power is not only interesting for ecological reasons, but also to reduce electrical bill costs. Svanfeldt-Winter et al. relate that 57% of monthly data-center costs are spent on servers' electricity bills [36].

As designers work towards faster systems there is an increasing concern about the expected power consumption of future systems. For exascale computing performance to be achieved in the years to come, energy efficiency will be the most important constraining factor [9,11]. This is why power consumption dominates every discussion of exascale [41], and in fact, there is still no clear roadmap to power-efficient exascale computing [19].

The technical report “The landscape of parallel computing: A view from Berkeley” [2], analyzes both extremes of computing: servers and embedded systems, and concludes that these extremes are “colliding and merging worlds”. One example is power consumption: until some years ago no one was concerned with power consumption in data centers, but today, most data centers are looking for energy efficient systems. Embedded systems, on the other hand, have always been concerned with battery life, requiring minimal power consumption. In fact, according to Jensen and Rodrigues, high performance systems must borrow concepts from embedded systems to achieve exascale performance [19].

In this work, we could verify that the use of low power servers in parallel applications is suitable. In our experiments, while power consumption goes down, performance stays almost the same for x86 based architectures typically used to build clusters. We find out that processing clusters based on low power systems such as ARM processors are feasible and that this is a nice way to decrease power usage of several server applications.

1.1. Contributions

Grounded in experimental analysis, our main contributions are:

* Corresponding author.

E-mail addresses: rafaelaroaca@ieee.org (R.V. Aroca), lmargos@computer.org (L.M.G. Gonçalves).

Glossary

AC	Alternating Current
ARM	Advanced RISC Machines. A RISC computer architecture widely used in embedded systems
CISC	Complex Instruction Set Computer
CPU	Central Processing Unit
DC	Direct Current
DDR	Double Data Rate
DSP	Digital Signal Processor
Exascale	Exascale computing are systems with floating point performance on the scale of 10^{18}
GPU	Graphical Processing Unit
GREEN500	List of top 500 most energy efficient computers
HPC	High Performance Computing
HTTP	Hyper Text Transfer Protocol
IT	Information Technology
LCD	Liquid Crystal Display
MFlop	A million floating point operations
MFlops	A million floating point operations per second
MFlops/W	The relation of how many Flops a computer can perform consuming one watt of energy
RISC	Restricted Instruction Set Computer
SIMD	Single Instruction Multiple Data
SoC	System on Chip
SQL	Structured Query Language. A standard language used to manipulate and search data in databases
TOP500	List of the 500 fastest supercomputers in the world
TrueRMS	True Root Mean Square. A technique used to accurately measure alternating current
W (Watt)	A measurement of power consumption. In this article we measured Direct Current values and calculated power using $P(W) = U(V) * I(A)$, where U is the tension in Volts and I is the current in Amperes. To measure Watts in alternating current we used $P(W) = U(V) * I(A) * f$, where f is the power factor correction

- A low cost and reliable testing setup to analyze servers' power efficiency;
- A systematic and quantitative evaluation of several server architectures running web server, database server and Linpack benchmarks.

1.2. Organization

This paper is structured as follows. Section 2 presents several differences of the analyzed systems and a review of the state of the art. Section 3 describes the equipment used in the measurements and our experimental setup, while Section 4 depicts the experimental results. Section 5 discusses the results addressing our main contributions and Section 6 features our final remarks.

2. Theory

This section discusses several differences of ARM and x86 architectures from theoretical and practical points of view, focusing on power efficiency aspects of these devices.

2.1. Architectural differences

ARM is a Restricted Instruction Set Computer (RISC), while x86 is a Complex Instruction Set Computer (CISC). RISC processors are

designed to work with simple and fixed length instructions (ideally all instructions have the same size and execute in a single clock cycle) while CISC machines have a variety of complex instructions of variable length. As RISC instructions sizes are fixed, fetching is simpler and both opcode and operand can be simultaneously accessed (because they are in a known memory position), thus simplifying the design of the control unit and requiring less power [35].

Besides dealing with variable-length instructions, CISC processors also have to deal with several addressing modes and formats while fetching and decoding instructions [18]. Moreover, although several powerful instructions are available in x86 CPUs, they are frequently unused. Jamil stresses that only 25% of CISC instructions are used 95% of the time [18].

As RISC instructions are simpler, more instructions are needed to perform the same function of fewer CISC instructions, causing RISC programs to be bigger. This was a problem in legacy computers with limited memory, but this is not a significant issue nowadays. Also, because RISC instructions execute in one clock cycle, the interrupt latency is lower, giving better overall system response times.

Today's CISC designs typically use a simple processor at its core. The core processor takes instructions called microcodes while a translator unit receives CISC commands and converts them into microcodes, which are similar to RISC instructions, making them hybrid RISC/CISC architectures [18]. Although flexible, this approach adds overhead [17], making the processor more complex and power hungry.

CISC systems use out-of-order execution to obtain better performance while typical RISC processors use in-order execution, which also consumes less power. According to Hoffman and Hedge, the Atom processor also uses in-order execution [15]. In relation to the evaluated devices, ARM Cortex-A8 and Cortex-A9 are similar, but Cortex-A8 uses in-order execution while ARM Cortex-A9 uses out-of-order execution.

2.2. Packaging and number of chips

Another difference is the number of chips needed to build a computer. ARM is both a RISC architecture and a company. This company does not build chips, but licenses their designs to several semiconductor companies that build processors with ARM cores. These companies frequently mix ARM processor cores and other features into a single chip to build Systems on Chip (SoCs). SoCs are mainly used in phones and other consumer electronics devices so that a product can be built with less parts. NVIDIA, for example, offers the Tegra family that includes multi-core ARM processors and high performance Graphical Processing Units (GPUs) in a single chip. All the ARM processors analyzed in this work are actually SoCs with ARM cores.

A typical x86 computer needs several auxiliary chips in order to work. As shown in Fig. 1, x86 processors are connected to auxiliary chips called "bridges" that connect the processors to several other devices. In some systems, the x86 processor accesses the main memory and video via a bridge, while recent designs gave the processor direct I/O to memory and GPUs as shown by the dashed lines of Fig. 1. Reddi et al. explain that although Atom typically uses 1.5 to 4.5 W of energy, other companion components in an x86 computer draw from 20 to 30 W of energy. They suggest that better integration and reduction of power due to peripheral parts is essential in the future [32]. As a promising solution, Intel already has a roadmap of Atom based SoCs [1,34,3].

Typical SoCs, on the other side, pack several functions into a single chip. As Fig. 1 shows, a SoC can have multi-core CPUs, auxiliary Digital Signal Processors (DSPs), auxiliary GPUs, memory, memory controllers, I/O ports and other features, all in a single

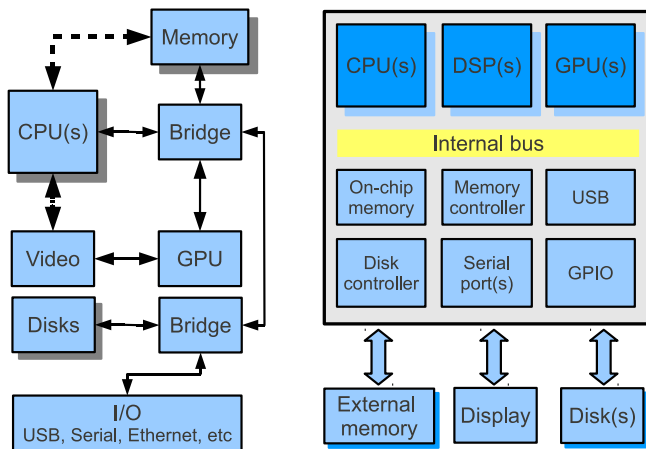


Fig. 1. A typical x86 based system (left) and a typical System On Chip based system (right).

chip. This leads to lower power consumption, requires less space in the circuit board and reduces external buses, allowing a system to be built with fewer components, reducing costs, interconnects, and the size of the final product.

The studied platforms are part of the Open Multimedia Application Processors (OMAP) SoCs. BeagleBoard-XM, with the OMAP3730 SoC, packs a Cortex-A8 ARM core, a Single Instruction Multiple Data (SIMD) unit for parallel fixed and floating point operations (128-bit), a GPU and a DSP. PandaBoard, with the OMAP4430 SoC, includes a dual core ARM Cortex-A9 processor, a GPU, a DSP and two more ARM Cortex-M3 processors. Both of these SoCs also have memory controllers, interface to external DDR memory and multiple I/O ports. For multimedia applications, for example, audio and video can be decoded in the DSP, leaving the processor free for other tasks. This allows it to be more responsive to user interactions and to run at slower speeds, consuming less energy.

2.3. Parallel execution and power management

Ideally, servers would use “energy proportional” systems to allow the power consumption to be directly related to the required performance, but unfortunately this does not happen in modern servers [36]. Another problem discussed by Svanfeldt-Winter et al. is that the best efficiency is at the server peak performance, but typical servers rarely reach such situations [36].

One of the advantages of using parallel computing is that it consists of an energy-efficient way of achieving performance [5], thus one important trend is the use of parallel architectures, even in embedded devices. A study shows that rendering a web page using a dual-core system with both cores running at half of their full speed has the same performance as using only one core at full speed [28], but with the difference that the solution using both cores draws 40% less energy.

According to Gruber and Keller, energy consumption is proportional to the frequency and to the square of the voltage (V) applied to the processor [12]. Poslad simplifies this relation and states that the power drawn by a CMOS processor can be approximated to V^3 . As the CPU frequency is proportional to the voltage, reducing the voltage causes greater power consumption reductions. This is consistent with the study mentioned above [28], where a single core at full speed consumes 1.1 V, with an approximate power of 1.33 W, while the same device with two cores at half speed consumes 0.8 V, with an approximate power of 0.51 W, which is 37% less than the power used by the single core at full speed.

CPUs consume 30–50% of a system’s total energy [30,41,14,36], so power management is crucial for embedded and mobile systems [30]. ARM architectures have evolved over many years taking energy requirements into consideration [36], while x86 processor companies were more focused on increasing performance and keeping compatibility with previous x86 systems. Some techniques commonly used in ARM devices such as Dynamic Voltage and Frequency Scaling (DVFS), have been included in x86 processors just in recent years. Such kinds of technology are now used in both architectures, allowing the operating system to decrease the clock rate, and consequently the power usage when the system load is low. DVFS in ARM devices typically have more frequency steps. ARM Cortex-A9, for example, has 9 possible power modes going from 1 GHz down to 200 MHz plus a deep sleep mode. This last mode, commonly used in phones, could be interesting for energy-scalable clusters. x86 computers, on the other hand, usually have less frequencies to choose from. In the tested systems, for example, the slowest possible frequency is 800 MHz for Turion, 1 GHz for Atom and 1.1 GHz for Xeon.

2.4. Low power servers and clusters

As traditional supercomputers consume large amounts of energy [14], the roadmap to exascale computing needs to include power usage reductions. Classical computer clusters need large spaces, so another issue is related to the lack of physical space, which is already a problem [16]. A SoC based cluster would certainly occupy less space (each board with 4 cores is less than 10×10 cm) and the cooling requirements would be less critical, reducing the costs with air conditioning. Moreover, such systems could be assembled without mechanical and moving parts, avoiding the use of mechanical hard disk drives and fans.

For ARM based systems, there is a lack of software support to fully use the potential and energy savings of such systems. This presents interesting research opportunities. Most benchmarks done with OMAP4430, for example, only consider its dual ARM Cortex-A9 cores, but better performance and efficiency could be achieved if OMAP4430’s DSP, GPU and the other two ARM Cortex-M3 cores were used. These parts were not used in our evaluations as in the tests done by other authors [9,15].

One of the first ARM based clusters was developed by Sandia National Laboratories with 196 cores. A company called Calxeda is shipping an ARM based platform specific for servers. Preliminary information from the company says that a system with 480 cores will consume 600 W of energy and offer the same performance as an equivalent x86 system that consumes 4000 W of energy. One of their products features a server with 120 ARM Cortex-A9 quad-core chips and a fast interconnect network [10]. One of the first proofs of concept of ARM Cortex clusters was created by German researchers [10]. Their 6-node system consumes 10 W of energy when highly loaded, with 16 MFlops/Watt performance [10].

ZT Systems offers an ARM based server with 16 cores consuming less than 80 W of energy. HP, a server manufacturer, has also put ARM based servers in its roadmap [27]. Marvell, another ARM chip maker, has announced a quad-core server SoC design based on the Cortex-A9 with support for DDR3 ECC memory and PCI-Express 2.0 interface directed to the server market [10]. ARM Holdings itself announced the ARM Cortex-A15 with several server features, including virtualization.

On the x86 side, SeaMicro sells a low power server with 512 Atom processors, reducing system costs by 75%. SeaMicro’s system also consumes about 3.5 times less energy at full load when compared to other similar servers [37]. They also offer a version with 768 Atom processors, which spends 18% less energy than the latter [26].

Table 1
Main specifications of the tested devices.

Device	Processor model	CPU clock (GHz)	Memory		
			RAM (MB)	Cache	Disk
Acer Notebook	AMD Turion MK-38 (1 core)	0.8	512	512 kB	USB Flash drive (2 GB)
Asus Notebook	Intel Atom N280 (1 core)	1	512	512 kB	USB Flash drive (2 GB)
HP-Z200	Intel Xeon X3450 (4 cores)	1.1	512	8 MB	USB Flash drive (2 GB)
PandaBoard (T.I. OMAP4430)	ARM Cortex-A9 (2 cores)	1	512	1 MB	SD Card (2 GB)
BeagleBoard-XM (T.I. OMAP3730)	ARM Cortex-A8 (1 core)	1	512	256 kB	Micro-SD Card (2 GB)

(T.I. = Texas Instruments).

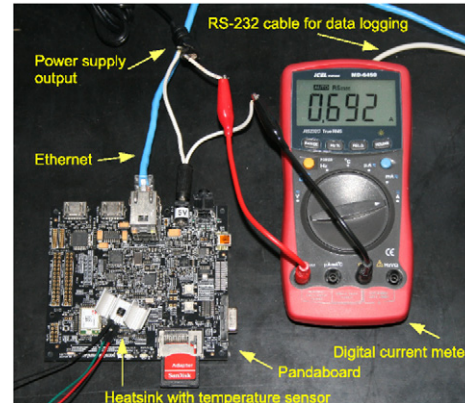
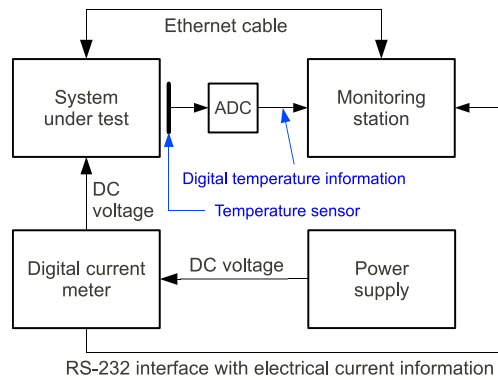


Fig. 2. Test system overview and PandaBoard under test.

Mont Blanc, a project that started in 2011, has the objective of building a scalable and power efficient HPC platform based on commercially available low-power embedded technology [24]. Other objectives includes the aim of reaching performances similar to the leaders of the TOP500 supercomputers list by 2017 and to the leaders of the GREEN500 list by 2014 [31,23]. To reach this goal, they intend to achieve 7 GFlops per Watt efficiency using ARM processors and low power GPUs [31]. As a result, this hybrid HPC design also promises to operate with 15–30 times less power consumption when compared to traditional HPCs [23–25].

As discussed in this section, the architecture of a system is an important and key part of its performance and power consumption. In that way, studying the architecture is important to analyze its performance and power efficiency, and also to take better advantage of efficient resources provided by the hardware, but that are not always used without explicit calls from the software.

3. Material and methods

3.1. Hardware setup

The hardware used for comparison purpose is composed of 2 notebook computers with low power x86 processors and 2 ARM based development boards. Table 1 shows the specifications for each device. We used two open source boards that can be easily acquired: the BeagleBoard-XM and PandaBoard. As a reference for the measurements, we also tested a classical machine used as a server in many scenarios. It is a HP-Z200 workstation with a quad-core Intel Xeon processor.

Platforms Turion and Atom have a dual channel DDR2 (Dual Data Rate) memory interface with 667 MHz bus clock (for Turion, both are used, but for Atom, the notebook only has one memory socket, so only one channel is used) while OMAP4430 features dual channel low power DDR2 memories with a 400 MHz bus clock. The system based on OMAP3730 has a single channel low power DDR memory interface with a 166 MHz bus clock. The Xeon based system has DDR3 memories with a 1333 MHz dual channel bus clock (all channels are used) and ECC (Error Correcting Code).

As each machine listed in Table 1 has specific features, some care should be taken in order to get a more fair performance comparison. First, we configure all devices to run at a similar processor clock speed. Although some of the tested devices can achieve higher speeds, such as the Xeon that can go up to about 3 GHz, the fastest speed of ARM is 1 GHz. For this reason, we configure the systems to be tested with similar speeds and thus making the comparison more fair. We also turn off the hard disks, wifi, Bluetooth and LCD display panels of the tested systems. For the laptop systems, we make the tests running the computers from the battery and the data is collected from the /proc interface provided by Linux's kernel. The collected data are electrical current, electrical voltage, temperature and CPU load.

To analyze the ARM systems we use a digital multimeter (Icel MD-6450 with RS-232 interface) with its inputs serially connected with the Direct Current (DC) output of the power supply as shown in Fig. 2. The multimeter measures the current from the boards and sends it directly to the monitoring station. The voltage is obtained at the start of each test run and these values are applied to the power equation for DC circuits ($P = V * I$), where P is the power in Watts, V is the tension in Volts and I is the current in Amperes. This setup is similar to the one used by Reddi et al. to measure web server power efficiency [32]. Fig. 2 shows a block diagram of the testing environment. We remark that this testing architecture can be easily reproduced, the built setup also being a contribution of our work.

To complete the hardware setup, we install passive heatsinks in the ARM boards. Although they typically do not use heatsinks, we want to keep all devices running the tests in similar conditions. Even though all tested x86 processors have heatsinks with active cooling (fans) while the ones used in the ARMs do not have.

The temperature measurements in the notebooks and Xeon are taken using their built-in CPU temperature sensor. As BeagleBoard's processor does not have a temperature sensor, we glue a calibrated temperature sensor to its heatsink. The same setup is applied to PandaBoard. As shown in Fig. 2, the temperature sensor is connected to a 10-bit analog to digital converter (ADC)

and the temperature measurements are automatically collected by the monitoring station.

There is an exception in the described method for the power measurements in the Xeon based workstation: it can be easily applied to a DC power supply with only one output, but workstations such as the HP-Z200 have several DC voltage outputs, making it difficult to measure all different tensions going out from the power supply. Our solution was to directly measure the current from the Alternating Current (AC) power cord before the computer's power supply. We remark that our results are accurate because the device's power supply uses active power factor correction, which makes the power factor to be near unity [33]. As we are dealing with AC current, we use the equation $P = V * I * f$ to calculate power considering the power factor correction (f). We also choose to use a TrueRMS digital multimeter to precisely measure AC current. Moreover, the power supply used in the experiments has an efficiency of 89%, which is considered in computing the final power consumption.

As the monitoring station, a Intel Core 2 Duo computer with 4 GB RAM memory and 100 Mbit fast ethernet network running Ubuntu Linux was used.

3.2. Software setup

All systems use Linux operating system with kernels 2.6.(32,35, 37). The database server used in all systems is MySQL 5.1 and the web server is Apache HTTP server version 2.2.16. As shown in Table 1, all systems were booted from USB or SD card flash drives and the tests were executed with data from RAM memory to avoid disk Input/Output (I/O) wait interference in the measurements. We used the Apache HTTP webserver default configuration, modifying it to support up to 500 concurrent threads (one per client connection).

In order to run the tests and collect data, a remote monitoring station controls the tests using remote shell commands. As was said, the systems are configured to run with a fixed CPU clock of about 1 GHz and the RAM memory of all systems is limited at boot to 512 MB (although they have bigger memories). Running programs are kept to minimal as possible and graphical user interfaces are not used.

To test the database we use the `mysqlslap` tool, which is a diagnostic program designed to emulate client load for a MySQL server. To test the web server we use the Apache benchmark (`ab`) tool. Both tools are executed in the monitoring host, which sends requests to the system under test using a direct 100 Mbit Ethernet cable between the machines as shown in Fig. 2. To test the floating point performance, we use the C version of Linpack [40]. The test algorithm is:

1. If it is a database server test, delete all data and tables and create new test tables with default values;
2. Reset power, temperature and CPU counters;
3. Start logging temperature and power usage readings;
4. Start logging CPU usage and I/O wait;
5. Start logging how many requests per second the server handles;
6. Send N simultaneous test requests (SQL or HTTP) to the system under test, measuring response time of each request (30 interactions each test);
7. If it is a floating point test, run Linpack in a $N \times N$ system of matrices;
8. Compute the average value for all logged measurements;
9. If it is a Linpack test, collect the obtained performance;
10. Delay 20 seconds to isolate interference from one test to another (power usage and CPU load, for example);
11. Restart in step 2 with the next N (number of concurrent clients or Linpack problem size).

As described in the algorithm, in order to achieve a better experimental validation, we performed each test separately from the others. For each test, the tools generated a sequence of 10,000 HTTP requests or 512 SQL requests. During the test execution, the monitoring station collected temperature, CPU usage, I/O wait and power usage at each second. For the Linpack test, the specified value N is used to perform benchmarks of LU factorization in a system of $N \times N$ matrices.

The HTTP server test consisted of downloading a simple static web page of 3145 bytes from the server, which is an embarrassingly parallel problem (there's little or no dependency between the concurrent tasks) [8]. The database server test, on the other hand, has some interactions between threads due to MySQL table locks. Before each test, a table is created with a numerical primary key (auto increment) and a timestamp field, then 2048 rows are populated with random timestamp values. Each test consists of computing the time difference from the value of each row to the current timestamp and summing up all results with a SQL aggregate function. This type of query avoids the database to cache results. The exact SQL command issued by each test query is "SELECT sum(now()-test.data) FROM test;".

For the tests with Linpack, we compiled the Linpack C version with the gcc compiler. For the ARM processors, we explicitly used a compiler optimization option to instruct the compiler to use the NEON Single Instruction Multiple Data (SIMD) unit (`gcc -mfp=neon`). NEON is capable of doing math operations with up to 4 floating point values simultaneously.

During the first tests, we noticed some problems and error messages related to network connections. In order to better execute the tests, we tuned several parameters of Linux's kernel to optimize the network and the TCP/IP stack for our test purposes. In all devices, we changed kernel options to support 64 000 open files and 32 000 network connections. We enabled `tcp_tw_reuse` and `tcp_tw_recycle` features, disabled the option `tcp_syncookies` and decreased the TCP/IP FIN timeout (`tcp_fin_timeout`) to 10 s.

We also measured latencies for each test. The motivation for measuring response time aims to verify the impact of low power processors on overall service quality, as some authors argue that these processors can impact the system's quality of service when tasks require high workloads [32].

4. Results

This section presents the results obtained from our measurements. As explained in Section 3 the systems were prepared to have minimal or no disk I/O during the tests. I/O wait was also collected in parallel to every measurement that is executed, but we decided to omit all I/O wait values in the graphs because the worst measured I/O wait is 2.8%. All the tests were executed according to the measurement technique presented in Section 3.

Another observation regards network bandwidth. A series of file transfers between the monitoring machine and all devices under test were executed in order to check the network bandwidth, which is 11 MB/s. As the test that uses more bandwidth does not exceed 7 MB/s, we can ensure that network bandwidth is not a bottleneck during the tests.

4.1. Web server

Fig. 3 shows plots of the average values collected in all platforms during the web server evaluation. Tests are executed ranging from 1 to 1000 simultaneous clients concurrently requesting static web pages with 25 clients interval steps (1, 25, 50, . . . , 1000). The plots are smoothed using a Bezier curve approximation. In the plots of Fig. 3, solid lines show the results for each tested system with all available cores enabled, while dashed lines show results for each

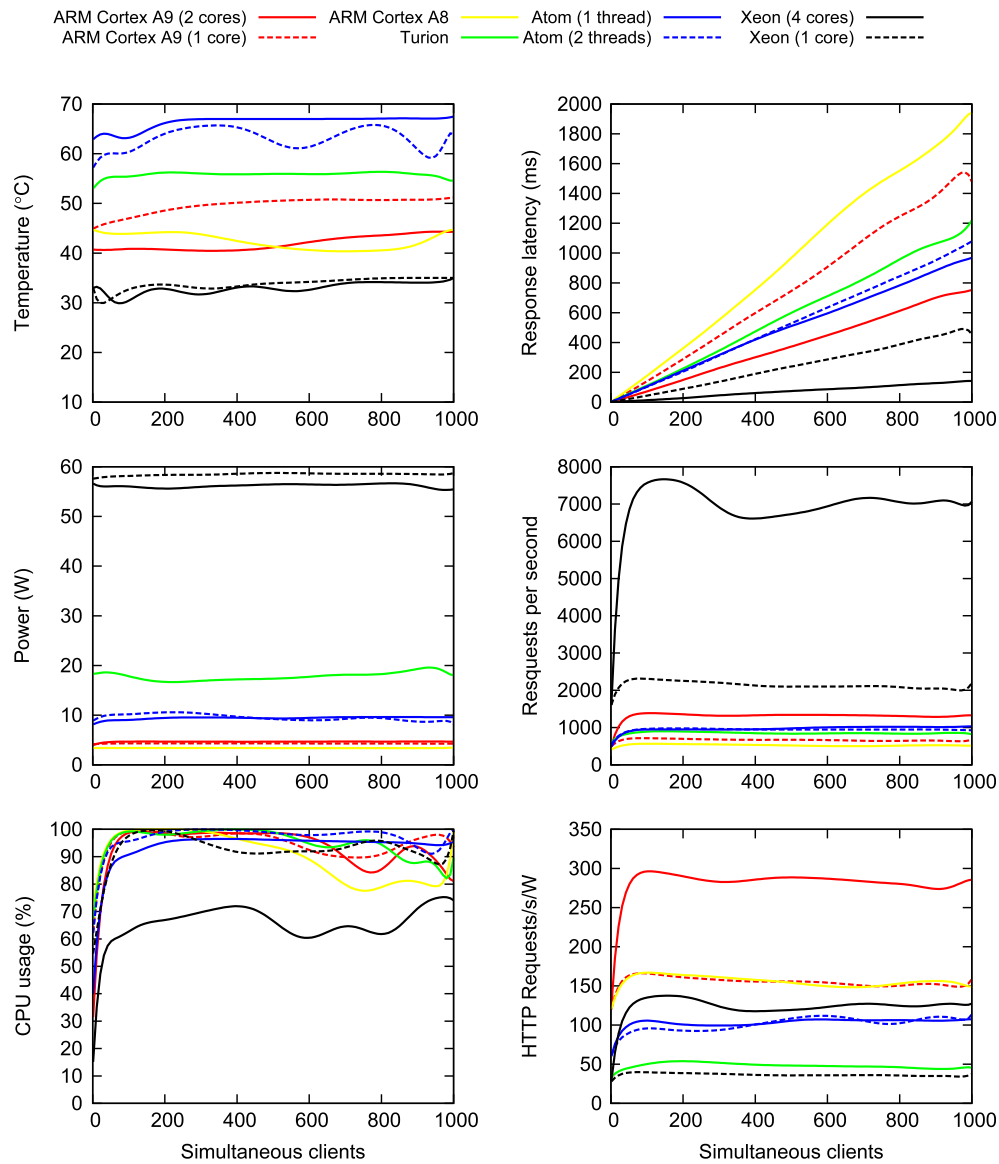


Fig. 3. Average values for Apache HTTP web server measurements in all platforms.

device with only one core enabled. The Atom processor is single core with HyperThreading technology, so we also tested it with HyperThreading enabled and disabled.

The temperature graph shows one interesting fact about the Xeon machine: it is the coolest tested system. This happens due to two factors: the first is because the Xeon processor is underclocked to 1.1 GHz, but its cooling system is designed to cool the processor at speeds of 3 GHz. Moreover, the HP-Z200 workstation has a robust cooling system with a big heatsink and a cooling fan. At the other extreme of the graph, the Atom notebook had only a small cooling fan and a heatsink fixed to the notebook's chassis. For the ARM Cortex-A8, we could verify that the temperature decreases when the processor usage and power consumption also decrease. The Turion platform maintains a constant value for the temperature that decreases a little when the processor usage and power consumption also decrease. Both Xeon and Cortex-A9 show hotter temperatures when only one core is used, which makes sense with the relation described in Section 2.3, because only one core needs more energy, and dissipates more power.

The power graph shows the energy usage for each concurrent number of clients. The average power consumption (over all number of clients tests) for Atom is 8.98 W with two threads and

9.32 W with two threads. Turion average power consumption is 17.38 W with two threads. ARM Cortex-A8 is only 3.29 W. ARM Cortex-A9 results are 4 W with only one core and 4.54 W with dual core. With only one core, the Xeon average is 64.13 W and 61.60 W with 4 cores running. Reddi et al. have measured a similar value, 62.5 W, for a Xeon based system running Microsoft Bing web search service. On their measurements, Atom consumed only 3.2 W of energy, about 1/3 of the values that we have measured.

The CPU usage graph shows that our tests keep the CPUs under heavy load most of the time, except for the Xeon with 4 cores, which, in this case, has much more processing power than necessary. Other system bottlenecks can also be the cause, such as if memory could not deliver information fast enough for the processors to work.

The response latency graph gives a clue of the quality of service provided by each device. As expected, Xeon has the best response times, with 70 ms average and 146 ms maximum response times using 4 cores. With one core, the maximum latency is 549 ms with 234 ms average. ARM Cortex-A8 has a 942 ms average and 1943 ms worst response time when 1000 clients are doing requests, which are unacceptable response times for interactive web applications. Turion worst time is 1224 ms and average time is 577 ms. Atom

average value (with HyperThreading) is 516 ms and its worst value is 969 ms. ARM Cortex-A9 has the second best performance with two cores, with a worst value of 753 ms and average of 370 ms. With only one core, the worst response time increases to 1667 ms.

It is interesting to note that the obtained rank seems related to the memory hierarchy quality. Xeon, which is the machine with the best results, has the fastest memory bus with dual channel and 8 MB cache memory. The second best device, ARM Cortex-A9, also has a dual channel memory system, but with a slower memory bus and 1 MB of cache memory. Atom, the third machine in the rank, has a faster memory interface than Cortex-A9, but only 512 kB of cache memory. Turion also has 512 kB of cache memory and the same memory speed as the Atom device. Last in the rank, Cortex-A8 has the lowest memory bus speed and lowest cache size (256 kB). Svanfeldt-Winter et al. [36] found an interesting issue during performance tests with Xeon systems. While trying to stress a Xeon test server, they were not able to obtain 100% of CPU utilization with a quad-core Xeon (as we also could not). Their investigation points out that neither the CPU nor the network are the bottlenecks, leaving the conclusion that memory is the bottleneck. They also argue that this happens because of “unbalanced components in a server”, suggesting the use of less powerful processors to solve this problem.

As this test does not have inter-process concurrency and locking mechanisms, it is expected that performance results scale up with the number of enabled processor cores. This can be clearly seen by analyzing the “Request per second graph”, where Xeon with 4 cores handles roughly 4 times more requests than Xeon with only one core (from 2086 to 6787 requests per second averages with peaks of 7813 for 4 cores and 2468 for one core). The ARM Cortex-A9 has a peak of 1448 requests per second with 2 cores and 750 with one core (averages of 1284 and 621 respectively). The Atom average value is 926 and the peak value is 1044 requests per second. Turion peak is 956 and average is 831. ARM Cortex-A8 top value is 600 requests per second with an average of 514.

Finally, the plot HTTP Requests/s/Watt is a direct division of the sustained requests per second by the consumed power, for each number of concurrent clients. This graph clearly shows that all ARM based devices have a better requests/s per Watt efficiency, even under heavy load, as expected from the analysis done in Section 2. Xeon running with a single core has the worst performance per Watt, followed by Turion and Atom. Interestingly, ARM Cortex-A8 and A9 have almost exactly the same performance per Watt with only one core. In general, ARM Cortex-A9 (with two cores) is 8.7 times more energy efficient than Xeon with only one core and 2.5 times more efficient than Xeon with 4 cores. Reddi et al., in their Microsoft web search application, have found that Atom is five times more efficient than Xeon, but query latencies on Atom are higher even with small load [32]. Our measurements show that Atom is 2–3 times more power efficient than Xeon with only one core, but also with higher latencies.

Svanfeldt-Winter et al. have executed a series of similar tests, but no direct measures of real power used by the systems are done in their research. In their work, tests of performance are done with an Apache HTTP server and the test results are related with the theoretical power usage of each device provided by the manufacturer. They found that the ARM Cortex-A9 can handle 3.6 times more traffic per Watt when compared to a Xeon machine [36]. In our experimental measurements, ARM Cortex-A9 can handle 2.5 more requests per Watt.

From the tests done for the webserver evaluation using static web pages, it can be noticed that the ARM based systems had a better performance per Watt ratio in all cases, but their response latencies are worse than the latencies of the Xeon based system, which has a faster memory bus and bigger cache memory. The cache size has also shown to be related to the response

performance. Moreover, the number of enabled cores in each system has also shown to be directly related to the obtained performance due to the fact that there is no dependency among tasks in the tested scenarios.

4.2. Database server

Fig. 4 shows plots of the average values collected in all platforms during the database server evaluation. Standard MySQL installation was used in all devices with a thread concurrency configuration parameter equal to the number of cores in each machine. Each plot of Fig. 4 shows results for the MySQL server running with clients from 1 to 130 with increases of 1 client at each measurement batch (1, 2, 3, ..., 130).

As shown in Fig. 4, the temperatures are similar to the temperatures of HTTP tests (Fig. 3). The hottest processor is Atom, followed by Turion, then ARM Cortex-A9, ARM Cortex-A8 and Xeon. An increase of temperature can be clearly seen between 0 and 80 simultaneous clients for Xeon and ARM Cortex-A9. There is also a notable increase of power usage and CPU usage in this interval.

The power plot shows that Xeon under heavy CPU load uses more power with four cores running than with a single core, which is expected. The peak power usage of Dual core Cortex-A9 is 5.34 W (average of 4.52 W) and 4.63 W peak power usage with one core (average of 4.09 W). Atom average power usage is 7.25 W with a peak of 7.41 W. Turion peak power is 18.1 W with an average of 14.62 W. ARM Cortex-A8 average power usage is 2.76 W with a peak of 2.94 W. Xeon used 62 W at peak with one core and 73.3 W with 4 cores (averages of 61.5 W and 67.89 W respectively).

In contrast to the HTTP graph of Fig. 3, CPU usage shown in Fig. 4 has many variations, but we can verify from the graph that all processors are under heavy load from 0 to 80 simultaneous SQL clients, each issuing 512 queries. The same trend can be seen in the plots showing the sustained requests per second and latency. A deeper analysis of this behavior is outside the scope of this paper, but it is very likely that they happen because of inter-process communications and Mutexes conflicts inside the database engine. Although this is an unexpected behavior because our tests execute only reading operations (SELECTs), this kind of issue seems to be known and even considered a bug in some MySQL versions. In fact, some database versions do use a global Mutex for each scanned row, resulting in a “Mutex ping-pong” problem when multiple CPUs are in use [42,29] causing systems with several CPU cores to execute concurrent queries slower than systems with only one core [39].

The described problem is consistent with the results of the “Response latency” and “Requests per second” plots. As can be seen in the plots, when more than 1 CPU core is used, the latency increases and the number of sustained requests per second decreases. For our test cases, unfortunately, parallel execution degrades performance.

The maximum latencies to answer the database queries are: Atom: 51.46 ms, Turion: 17 ms, Cortex-A8: 77 ms, Cortex-A9 (2 cores): 59.62 ms, Cortex-A9 (1 core): 46.96 ms, Xeon (1 core): 9.24 ms, and Xeon (4 cores): 8.57 ms. The average values are: Atom: 32.47 ms, Turion: 6.72 ms, Cortex-A8: 40.41 ms, Cortex-A9 (2 cores): 27.10 ms, Cortex-A9 (1 core): 25.24 ms, Xeon (1 core): 4.26 ms, and Xeon (4 cores): 4.16 ms.

The maximum sustained queries per second are (rounded): Atom: 50, Turion: 152, Cortex-A8: 41, Cortex-A9 (2 cores): 105, Cortex-A9 (1 core): 125, Xeon (1 core): 349, and Xeon (4 cores): 220. The average values are: Atom: 36, Turion: 57, Cortex-A8: 29, Cortex-A9 (2 cores): 49, Cortex-A9 (1 core): 56, Xeon (1 core): 116, and Xeon (4 cores): 112.

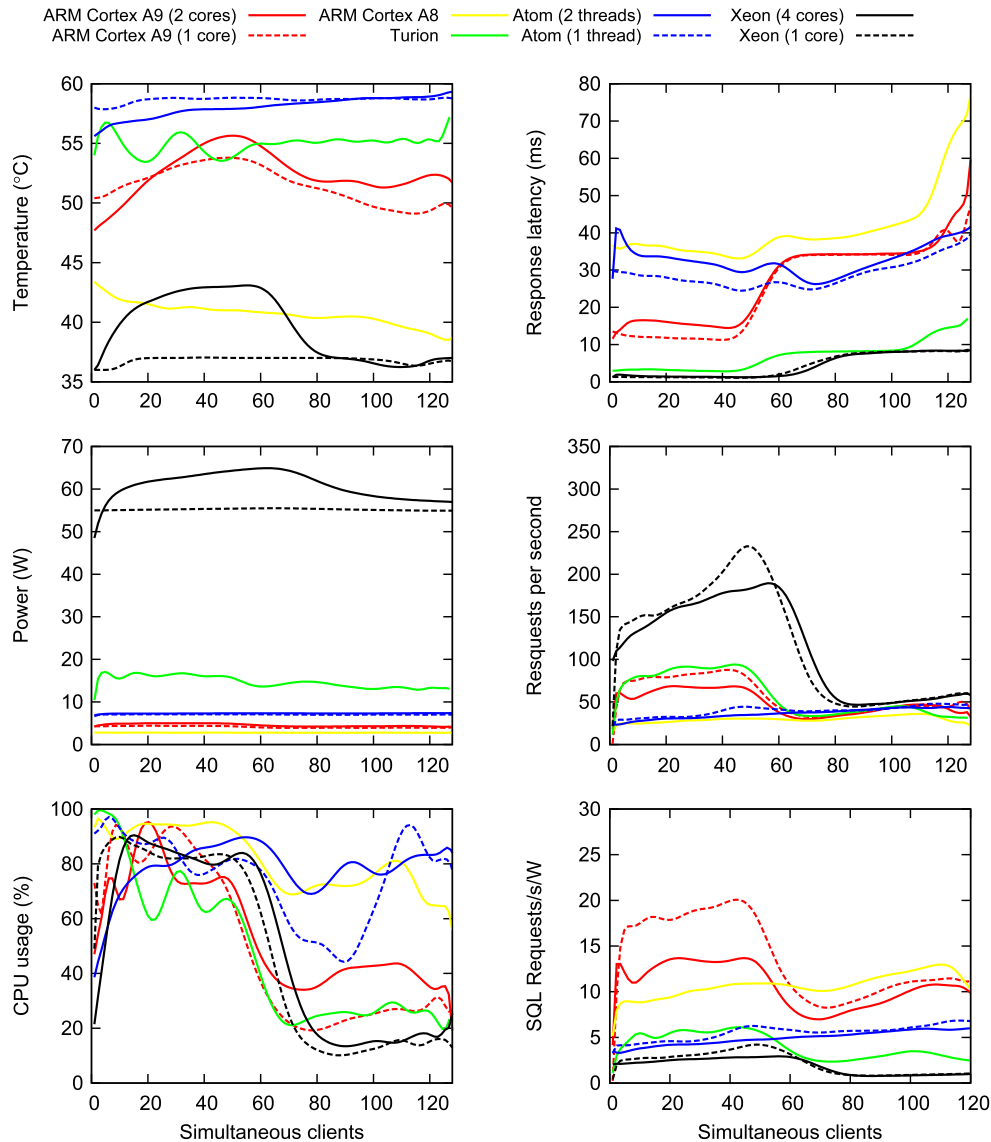


Fig. 4. Average values for MySQL database server measurements in all platforms.

As we already discussed, in this particular set of tests, better results are obtained using a single core, and from this fact, better power efficiency is obtained with the systems using only one core. This can be seen in the last plot of Fig. 4. Again, ARM Cortex-A9 is the most power efficient system among the tested ones, followed by ARM Cortex-A8, Turion, Atom, and Xeon at last.

From the tests done for the database server evaluation, the ARM based systems, as in the webserver tests, had a better performance per Watt ratio in all cases. Also, as in the webserver tests, their latencies were worst than the Xeon based system. Contrary to the HTTP server test, which had few inter-process communications, the database server had concurrency issues. As a consequence, unfortunately, better results were obtained when only one processor core was enabled. When more cores were activated, both absolute performance and performance per Watt decreased. This result clearly shows that parallel programming aspects such as concurrency and locks must be carefully addressed to achieve better power efficiency in multi-core systems.

4.3. Floating point

This section describes and discusses the tests executed to analyze floating point performance on all processors. In these tests,

only the serial version of Linpack is executed, so the test is done using only one core of each tested system with matrix sizes ranging from 20×20 to 2500×2500 . We have taken this opportunity to execute the tests at different CPU clock frequencies. Fig. 5 shows plots of the average values collected in all platforms during the tests. This figure also includes a plot with the summary of the results and the MFlops per Watt rates observed in each system.

The temperature graph shows that increasing clock speed, as it is known, increases temperature. We can note that the temperature increases from 10 to 15 degrees Celsius depending on the system. Turion is the hottest processor at 2.2 GHz, followed by Atom at 1.6 GHz and then Xeon at 2.67 GHz. As we already mentioned in Section 4.1, the Xeon cooling system is more robust than the others tested, keeping the system cooler.

The power graph shows that Turion at 2.2 GHz practically uses double the energy when compared to itself at 800 MHz. Xeon power usage increase is modest, because of the overall system's power consumption. The power usage for each system is: peak of 10 W for Atom at 1.6 GHz with average of 9.19 W, peak of 8.22 W for Atom at 1 GHz with average of 7.79 W, peak of 3.58 W for ARM Cortex-A8 at 1 GHz with average value of 3.44 W, peak of 4.54 W for ARM Cortex-A9 at 1 GHz with average value of 4.28 W, peak of 45.1 W for Turion at 2.2 GHz with average value of 39.24 W and

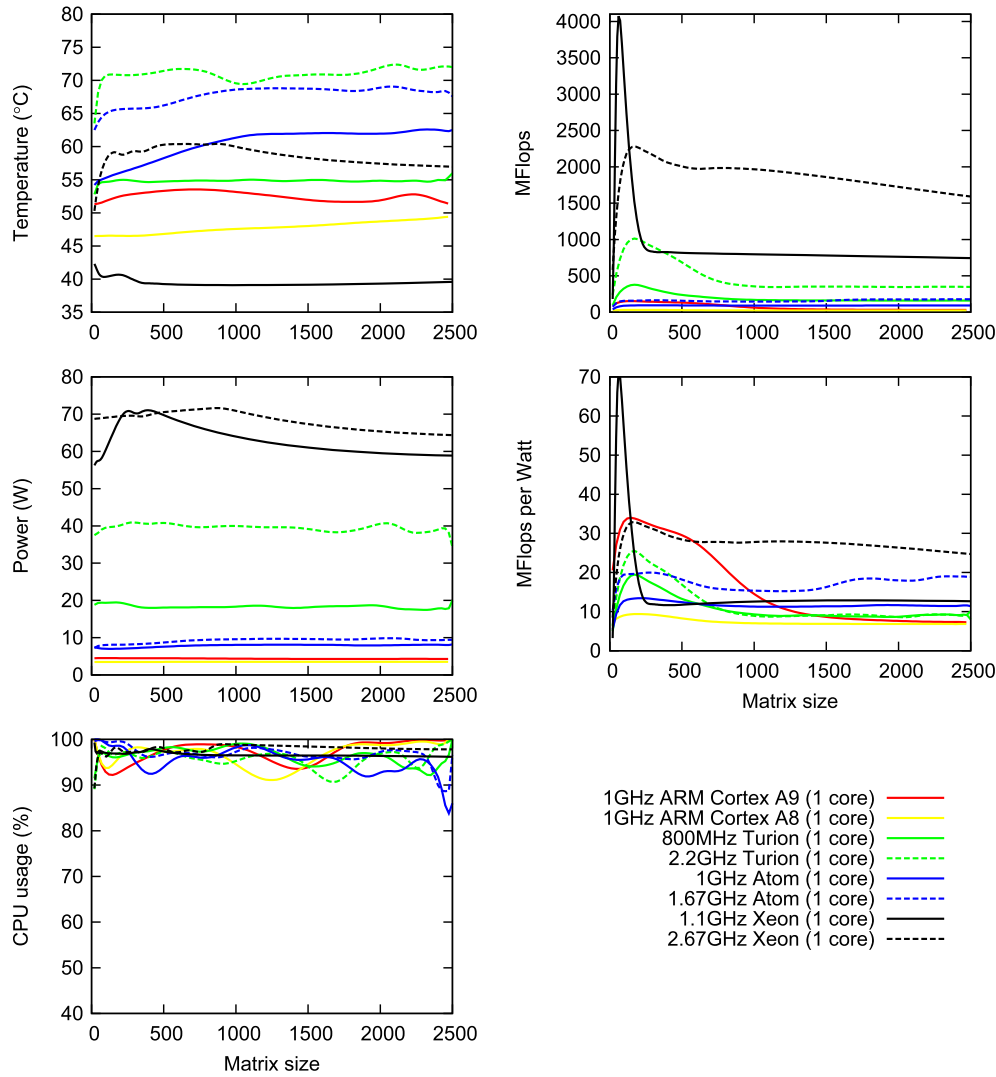


Fig. 5. Floating point results obtained from the measurements of Linpack executions.

peak of 20.1 W for Turion at 800 MHz with average of 18.15 W. Xeon maximum power usage is 80.8 W with an average of 71.18 W at 1.1 GHz and a maximum of 80.9 W with an average of 76.38 W at 2.67 GHz. CPU usage, as shown in the graph, is always above 90% for all processors (most of the time at 100%).

The MFlops graph shows the results collected after the execution of Linpack given each matrix size. All systems have peak performance while Linpack is executing the LU factorization for a system of matrices of sizes 40×40 , 50×50 , 60×60 , 70×70 and 80×80 . Xeon has the most prominent performance for these matrix sizes. As Linpack is a CPU bound process, the only bottlenecks are the CPU speed and memory I/O. If all data fit into the cache, then computation is much faster, which probably happens at peak values. As said already, Xeon has a memory system with better performance and the biggest cache of all tested systems. The results are: Atom (1.6 GHz) has a peak of 178 MFlops with an average of 158 MFlops and a peak of 98 MFlops with an average of 91 MFlops at 1 GHz. ARM Cortex-A8 has a peak of 33 MFlops with an average of 25 MFlops and Cortex-A9 has a peak of 172 MFlops with an average of 68 MFlops. Turion (2.2 GHz) has a peak of 1.1 GFlops with an average of 464 MFlops and a peak of 450 MFlops with an average of 194 MFlops at 800 MHz. Xeon has a peak of 6 GFlops with an average of 1 GFlop at 1.1 GHz and a peak of 2.4 GFlops with an average of 1.8 GFlops at 2.67 GHz.

Some of the values that we present here are similar to the ones that are measured by Hoffman and Hedge [15], such as the 23

MFlops for ARM Cortex-A8. Fuerlinger et al. measure performance in two different Cortex-A8 devices with different memories setup. Although with a small clock rate difference between their two measurements (one system at 1 GHz and another at 800 MHz), their data show that memory impact is significant: with a 200 MHz DDR2 64-bit bus memory, Linpack performance reaches 57.5 MFlops while the same system with a 166 MHz DDR 32-bit bus memory has a performance of less than half (22.6 MFlops). They also measure memory transfer speed and the system with 166 MHz memory has a speed of 481 MB/s while the one with 200 MHz has a transfer rate of 749.8 MB/s, 1.5 times faster [10].

According to the “MFlops per Watt” graph, Xeon at 2.67 GHz has the best power efficiency in the long run. Although ARM Cortex-A9 has better power efficiency for matrices between 20×20 and 500×500 , only Xeon keeps its performance constant with different problem sizes for all tested matrix sizes. This is consistent with results from other authors that have observed that “x86” computers are better when raw performance is considered [15].

Specifically about the floating point performance of ARM devices, we explicitly instructed the compiler to generate the Linpack executable to use the NEON SIMD unit as the floating point unit. When running all tests at 1 GHz clock, we note that, without NEON, Linpack performance results are from 2 to 3 times worse. Theoretically, the NEON unit of the tested devices could execute up to 4 single precision operations at the same time. Another

interesting fact is that Linpack using NEON, consuming 100% of CPU time, spends an average of 3.46 W while without NEON, with 3 times worse performance, the average energy used is 3.65 W, showing again that parallel programming is an energy-efficient way to achieve performance. Finally, as stated in Section 2.1, when NEON is not used, the executable program disassemble contains 4329 instructions while with NEON the program size is reduced to 2373 instructions, because single SIMD instructions can substitute for several simpler standard RISC instructions.

From the tests done for the floating point performance evaluation, all x86 systems had better absolute floating point performance. For these x86 systems, we could observe that, as expected, when the processor clock is increased, its temperature also increases considerably. As floating point is mainly a CPU-bound task, CPU speed and memory access speed are key to the resulting performance. Cache memory is also important. The ARM based systems, even using the NEON SIMD unit, had the worst performance, which would be even worse without NEON. The overall result is that raw floating point performance is still better on x86 processors.

5. Discussion

According to Cameron [4], one of the challenges of green IT is the lack of more efficient hardware and software. As an answer, our experiments show that it is viable to build clusters and servers of low power devices to act as energy efficient production systems.

One possible solution to decrease data centers power usage is to put x86 servers to sleep and wake them up with Wake On LAN as the computing demand increases. Although this approach would certainly reduce power usage, servers in sleep mode consume from 5 W to 15 W of energy. We verify that a BeagleBoard-XM with ARM Cortex-A8 running at 300 MHz consumes only 2.7 W of energy with the CPU at 100% of usage. So, the power required to keep one server in standby can roughly keep 2 ARM based systems running under full load. While the server in standby cannot do any job, an ARM based system can do a variety of jobs with the power that is simply lost to keep traditional servers in standby.

ARM is the most widely used 32-bit RISC architecture [7] in the world. According to one of ARM Holdings founders, it can do the same amount of work as other 32-bit processors do but consuming one-tenth of the energy [7]. Furthermore, there are hundreds of companies manufacturing ARM compatible processors, keeping the price low due to concurrency and giving designers the freedom to choose from several suppliers. Other authors say that, simpler processors, such as ARMs, can reduce high performance computing systems' power consumption by 95% [19].

On the high performance side, we have not explored additional features of ARM based SoCs in our tests, such as DSPs, GPUs and other asymmetrical cores available on chip. The DSP, for example, could be used in a variety of situations, such as handling secure socket layer (SSL) processing requests or other computations, helping to increase performance. In fact, a research company (IDC), argues that the fastest path to the exaflop milestone is through heterogeneous designs [13], such as using hybrid systems with ARMs, GPUs and DSPs. Moreover, 28% of HPC systems are already using some sort of auxiliary acceleration system to help x86 CPUs, such as GPUs [16].

While x86 manufacturers such as Intel are already working on low power SoCs such as Atom designs with integrated bridges (also called a chipset), several companies are delivering ARM based server solutions as described in Section 2.4. ARM powered systems are promising if they could have caches and memory interfaces similar to typical x86 servers. Marvell, another company that provides ARM based SoCs introduced Armada XP, a 1.6 GHz quad-core ARM system with 2 MB cache and 1.6 GHz 64-bit DDR2 or DDR3 memory interface with ECC. The system also has 4 gigabit ethernet ports, USB Host, SATA ports and PCI Express interfaces, all in the same chip [22] consuming less than 10 W of energy [38].

6. Conclusions

In this paper we present a low cost testing system to analyze computers' power efficiency that can be easily reproduced by other authors. Using this test setup we systematically analyze and compare several ARM and x86 devices in typical server and number crunching tasks: web server, database server and floating point computation. In the comparisons we show temperature, CPU and power usage for each device at different load scenarios. We also show performance per Watt metrics and the latency to serve HTTP or SQL requests, a good indicator of the quality of service offered by the system.

We notice that ARM based SoCs have a good performance to build servers and clusters, specially when we consider their performance per Watt relation. In the tests that we have done for HTTP and SQL servers, ARM devices are from 3 to 4 time more power efficient than x86 systems when we consider the requests per second per Watt relation under different load situations (from idle to heavily loaded). The exception is for floating point computation, where ARM Cortex-A9 had a superior efficiency only for a small number of problem sizes, and then its performance decreased. x86 processors, on the other hand, maintained their performance approximately at a constant rate, showing that x86 is still superior for floating point computation when both performance and power efficiency are considered.

While ARM processors, typically used in mobile and embedded systems, increasingly target servers, desktop and laptop computing, x86 processors, typically used in desktop and server computers, are constantly being redesigned to address the low power requirements of data centers and mobile devices. ARM based servers are already a reality in the same way that x86 Atom based phones are already a reality. Developers and designers need to rethink their software designs to take full advantage of these new devices powered with heterogeneous processing units.

Acknowledgment

The authors would like to thank the support from the National Research Council (CNPq), the Brazilian sponsoring Agency for research.

References

- [1] S. Anthony, Intel Medfield 32 nm Atom SoC power consumption, specs, and benchmarks leak, Extreme Tech. On-Line. Dec 17 2011. Access date: Mar/2012. URL: <http://www.extremetech.com/computing/110563-intel-medfield-32nm-atom-soc-power-consumption-specs-and-benchmarks-leak>.
- [2] K. Asanovic, R. Bodik, B.C. Catanzaro, J.J. Gebis, P. Husbands, K. Keutzer, D.A. Patterson, W.L. Plishker, J. Shalf, S.W. Williams, K.A. Yelick, The landscape of parallel computing research: a view from Berkeley, Technical Report UCB/EECS-2006-183, EECS Department, University of California, Berkeley (Dec 2006). On-Line. Access date: March 2012. URL: <http://www.eecs.berkeley.edu/Pubs/TechRpts/2006/EECS-2006-183.html>.
- [3] D. Athrow, Intel details roadmap, 14 nm Atom SoC To Come in 2014, IT Pro Portal. On-Line. Feb 28 2012. Access date: Mar/2012. URL: <http://www.itproportal.com/2012/02/28/intel-details-roadmap-14nm-atom-soc-come-2014/http://www.itproportal.com/2012/02/28/intel-details-roadmap-14nm-atom-soc-come-2014/>.
- [4] K.W. Cameron, Green with envy, Computer 43 (11) (2010) 95–96.
- [5] A.P. Chandrakasan, S. Sheng, R.W. Brodersen, Low-power CMOS digital design, solid-state circuits, IEEE Journal of Solid State Circuits 27 (1992) 473–484.
- [6] J. Dongarra, Performance of various computers using standard linear equations software, University of Tennessee, Knoxville TN, 37996, Computer Science Technical Report Number CS - 89 85. On-Line. Access date: Feb/2012. URL: <http://www.netlib.org/benchmark/performance.ps>.
- [7] J. Fitzpatrick, An interview with Steve Furber, Communications of ACM 54 (2011) 34–39.
- [8] I. Foster, Designing and Building Parallel Programs, first ed., Addison-Wesley, Boston, USA, 1995.
- [9] K. Fuerlinger, The AppleTV cluster: energy efficient parallel computing on consumer electronic devices, 12 August 2011. On-Line. Access date: March 2012. URL: <http://www.appletvcluster.com/>.

- [10] K. Fuerlinger, C. Klausecker, D. Kranzlmler, The AppleTV-Cluster: towards energy efficient parallel computing on consumer electronic devices. April 2011. Whitepaper v1.0, April 2011. On-Line. Access date: Mar/2012. URL: <http://www.nm.ifi.lmu.de/projects/ATV2CLUSTER/atvcluster.pdf>.
- [11] K. Fuerlinger, C. Klausecker, D. Kranzlmler, Towards energy efficient parallel computing on consumer electronic devices, in: ICT-GLOW 2011. Toulouse, France, September 2011.
- [12] R. Gruber, V. Keller, HPC@Green IT: Green High Performance Computing Methods, first ed., Springer, 2010.
- [13] S. Gupta, Path to exascale computing rests with heterogeneous design. Nvidia Corporation. On-Line. November 8 2011. Access date: Mar/2012. URL: <http://blogs.nvidia.com/2011/11/path-to-exascale-computing-rests-with-heterogeneous-design/>.
- [14] S. Gupta, Worlds first ARM-based supercomputer to launch in Barcelona. Nvidia Corporation. On-Line. November 14 2011. Access date: Mar/2012. URL: <http://blogs.nvidia.com/2011/11/worlds-first-arm-based-supercomputer-to-launch-in-barcelona/>.
- [15] K.R. Hoffman, P. Hedge, ARM Cortex-A8 vs. Intel Atom: architectural and benchmark comparisons. Technical Report. University of Texas at Dallas. 2009. On-Line. Access date: March 2012. URL: <http://caxapa.ru/thumbs/229665/armcortexa8vsintelatomarchitecturalandbe.pdf>.
- [16] IDC executive brief, heterogeneous computing: a new paradigm for the exascale era (Adapted from IDC HPC end-user study of processor and accelerator trends in technical computing by Earl C. Joseph and Steve Conway). November 2011. On-Line. Access date: March 2012. URL: http://5601-blogs-nvidia-com.voxcdn.com/wp-content/uploads/2011/11/IDC-Exascale-Executive-Brief_Nov2011.pdf.
- [17] C. Isen, L.K. John, E. John, A tale of two processors: revisiting the RISC-CISC debate, in: David Kaeli, Kai Sachs (Eds.), Proceedings of the 2009 SPEC Benchmark Workshop on Computer Performance Evaluation and Benchmarking, Springer-Verlag, Berlin, Heidelberg, 2009, pp. 57–76.
- [18] T. Jamil, RISC versus CISC: why less is more, IEEE Potentials 14 (1995) 13–16.
- [19] D. Jensen, A. Rodrigues, Embedded systems and exascale computing, Computing in Science Engineering 12 (2010) 20–29.
- [20] V. Kumar, A. Fedorova, Towards better performance per watt in virtual environments on asymmetric single-isa multi-core systems, ACM SIGOPS Operating Systems Review 43 (2009) 105–109.
- [21] P. Mahadevan, S. Banerjee, P. Sharma, A. Shah, P. Ranganathan, On energy efficiency for enterprise and data center networks, IEEE Communications Magazine 49 (2011) 94–100.
- [22] Marvell semiconductor, Marvell quad-Core ARMADA XP series SoC: for enterprise-class cloud computing applications. November 2010. On-Line. Access date: March 2012. URL: http://www.marvell.com/embedded-processors/armada-xp/assets/armada_xp_pb.pdf.
- [23] G. Millington, Barcelona supercomputing center to deploy world's first ARM-based CPU/GPU hybrid supercomputer — prototype system with energy efficient tegra ARM CPUs and CUDA GPUs advances Europe toward exascale supercomputing. NVIDIA Press Room, November 14 2011. On-Line. Access date: March 2012. URL: http://pressroom.nvidia.com/easyir/customrel.do?easyirid=A0D622CE9F579F09&version=live&spid=821220&releasejsp=release_157&html=true.
- [24] Mont Blanc website, Mont Blanc Project objectives. On-Line. 2011. Access date: March 2012. URL: <http://www.montblanc-project.eu/objectives>.
- [25] Mont Blanc website, Mont Blanc Project introduction. On-Line. 2011. Access date: March 2012. URL: <http://www.montblanc-project.eu/introduction>.
- [26] T.P. Morgan, SeaMicro pushes 'Atom smasher' to 768 cores in 10U box: A lot more bang for some more bucks. The Register, July 18 2011. On-Line. Access date: Mar/2012. URL: http://www.theregister.co.uk/2011/07/18/seamicro_sm10000_64hd_upgrade/.
- [27] T.P. Morgan, HP Project Moonshot hurls ARM servers into the heavens. The Register, November 1 2011. On-Line. Access date: Mar/2012. URL: http://www.theregister.co.uk/2011/11/01/hp_redstone_calxeda_servers/.
- [28] NVIDIA Corporation, Benefits of multiple cpu cores in mobile devices. White Paper. 2010. On-Line. Access date: March 2012. URL: http://www.nvidia.com/content/PDF/tegra_white_papers/Benefits-of-Multi-core-CPU-in-Mobile-Devices_Ver1.2.pdf.
- [29] Oracle Corporation, MySQL User's Guide, InnoDB performance and scalability enhancements, 7.14. Control of Spin Lock Polling. On-Line. Access date: March 2012. URL: http://dev.mysql.com/doc/innodb/1.1/en/innodb-performance-spin_lock_polling.html.
- [30] S. Poslad, Ubiquitous Computing: Smart Devices, Environments and Interactions, first ed., Wiley Publishing, New York, USA, 2009.
- [31] A. Ramirez, European scalable and power efficient HPC platform based on low-power embedded technology. On-Line. Access date: March/2012. URL: <http://www.eesi-project.eu/media/BarcelonaConference/Day2/13-Mont-Blanc-Overview.pdf>.
- [32] V.J. Reddi, B.C. Lee, T. Chilimbi, K. Vaid, Web search using mobile cores: quantifying and mitigating the price of efficiency, in: Proceedings of the 37th annual international symposium on Computer architecture, ISCA 2010, ACM, New York, NY, USA, 2010, pp. 314–325.
- [33] S. Sandler, Switchmode Power Supply Simulation with PSpice and SPICE 3, in: Electronic engineering series, McGraw-Hill, 2006.
- [34] A.L. Shimpi, Intel demonstrates dual-core Atom SoC with integrated WiFi transceiver, AnandTech. On-Line. Feb 20 2012. Access date: Mar/2012. URL: <http://www.anandtech.com/show/5554/intel-demonstrates-dualcore-atom-soc-with-integrated-wifi-transceiver>.
- [35] W. Stallings, Reduced instruction set computer architecture, Proceedings of the IEEE 75 (1988) 38–55.
- [36] O. Svanfeldt-Winter, S. Lafond, J. Lilius, Cost and energy reduction evaluation for ARM based web servers. IEEE Ninth International Conference on Dependable, Autonomic and Secure Computing, DASC, 12–14 Dec. 2011, pp. 480–487.
- [37] D. Takahashi, SeaMicro drops an atom bomb on the server industry June 13 2010. On-Line. Access date: March 2012. URL: <http://venturebeat.com/2010/06/13/seamicro-drops-an-atom-bomb-on-the-server-industry/>.
- [38] K. Thomas, Marvells Armada XP could bring a cloud revolution. November 9 2010. PC World. On-Line. Access date: March 2012. URL: http://www.pcworld.com/businesscenter/article/210156/marvells_armada_xp_could_bring_a_cloud_revolution.html.
- [39] V. Tkachenko, Returning to InnoDB scalability, July 2008. MySQL Performance Blog. On-Line. Access date: Mar/2012. URL: <http://www.mysqlperformanceblog.com/2006/07/28/returning-to-innodb-scalability/>.
- [40] B. Toy, LINPACK.C: linpack benchmark, calculates FLOPS (Floating Point Operations Per Second), May 1988. On-Line. Access date: March 2012. URL: Linpack from <http://www.netlib.org/benchmark/linpackc.new>.
- [41] T. Wilkie, From mobile phone to supercomputer? (Towards exascale), Scientific Computing World, 19 February 2012. On-Line. Access date: March 2012. URL: http://www.montblanc-project.eu/system/files/exascale_computing_article_02-2012.pdf.
- [42] P. Zaitsev, Very poor performance with multiple queries running concurrently. Bug #15815. 2005. On-Line. Access date: March 2012. URL: <http://bugs.mysql.com/bug.php?id=15815>.



Rafael Vidal Aroca holds an Informatics degree and a master of sciences degree in mechatronics engineering both from the University of São Paulo (USP). He is a Ph.D. student at the Electrical and Computing Engineering Graduate Program of Federal University of Rio Grande do Norte, Brazil. He has over 10 years of industry experience in embedded systems, IT systems and servers administration. His main research interests are in Embedded Systems, Operating Systems and Robotics. Rafael is an IEEE member.



Luiz Marcos Garcia Gonçalves holds a Doctorate in systems and computing engineering from the Federal University of Rio de Janeiro and graduated in 1999. He is Associate Professor at the Computing Engineering and Automation Department of the Federal University of Rio Grande do Norte, Brazil. He is a member of the IEEE Latin American Robotics Council (since 2002) and he was the Chair for the Brazilian Committee on Robotics and on Computer Graphics and Image Processing both under the Brazilian Computer Society. His research interests are in Computer Vision, Robotics, and all aspects of Graphics

Processing.